

Default Constructor

In Java, a **default constructor** is a constructor that is automatically provided by the Java compiler if no constructors are explicitly defined in the class. This default constructor initializes the object with default values. For reference, primitive fields are initialized to default values such as `0` for integers, `false` for booleans, and `null` for objects.

You can also **modify the object's fields** after it is created, either by directly accessing the fields (if they're public or if you have getter and setter methods) or through methods that modify the fields.

Here's an example of a Java program that:

Creates an object using the default constructor.

Code Example:

```
public class Car {
    // Fields of the Car class (instance variables)
    String make;
    String model;
    int year;

    // Default constructor (no parameters)
    public Car() {
        // The default constructor does not need to do anything explicitly
        // Fields will be initialized to their default values: null for
        Strings, 0 for int
    }

    // Method to display the current values of the object
    public void display() {
        System.out.println("Car make: " + make);
        System.out.println("Car model: " + model);
        System.out.println("Car year: " + year);
    }
}

public class Main {
    public static void main(String[] args) {
        // Create an object of the Car class using the default constructor
        Car myCar = new Car();

        // Display the default values of the object's fields
        System.out.println("modifying fields:");
    }
}
```

```
        myCar.display(); // Should show default values (null, null, 0)

    }
}
```

Explanation:

1. The Car class:

- It has three instance variables: `make`, `model`, and `year`.
- A **default constructor** is provided, which initializes the object with default values. For example, `make` and `model` are set to `null` (since they are `String`), and `year` is set to `0` (since it is an `int`).

2. The display method:

- This method is used to print the current values of the fields (`make`, `model`, `year`) to the console.

3. In the Main class:

- The `myCar` object is created using the default constructor, which initializes the fields with their default values.
- The `display` method is called to show the default values.

Output:

```
modifying fields:
Car make: null
Car model: null
Car year: 0
```

Key Points:

1. **Default Constructor:** If no constructor is explicitly provided, the Java compiler automatically provides a default constructor that initializes fields to their default values.
2. **Accessing Fields:** In this example, the `display` method is used to check the values of the object's fields.